

Differential CC $1\mu 1\pi^\pm$ cross sections on argon in the NuMI beam

Status of the blind analysis and a request to open the control regions

Ishanee Pophale and Jarek Nowak

Lancaster University

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$\nu_\mu + \bar{\nu}_\mu$ CC single charged pion on argon • FHC, RHC and combined • blind, fake-data results

What is measured

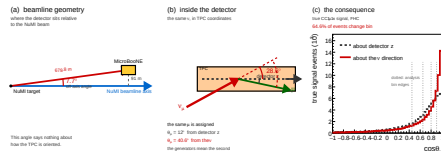
Signal (truth level, charge-blind): $\nu_\mu/\bar{\nu}_\mu$ CC in the FV with exactly one μ , exactly one $\pi^\pm$, no π^0 , no kaons or other mesons, any number of nucleons; coherent production included.

Measured phase space: $p_\mu > 0.15$, $p_\pi > 0.175$ GeV/c, $\theta_{\mu\pi} < 2.6$ rad

Deliverables per configuration (FHC, RHC, combined)

- ▶ total flux-averaged cross section (cut-and-count)
- ▶ $d\sigma/dp_\mu$, $d\sigma/d\cos\theta_\mu$ and $d\sigma/d\theta_\mu$, $d\sigma/d\cos\theta_\pi$, $d\sigma/d\theta_{\mu\pi}$, two-bin $d\sigma/dp_\pi$
- ▶ proton-tagged subsample: integrated cross section (headline); TKI and W_{had} as secondary results

Blind analysis: every “data” point in this talk is Poisson-thrown central-value MC. No beam-on signal region has been opened.



MicroBooNE sits 7.7° off the NuMI axis; the beam arrives 28° from detector z , so all angles and transverse quantities are computed in the beam frame.

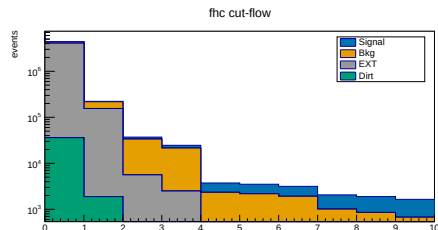
Selection and yields

Cuts: software trigger, vertex in FV, topological score > 0.67 , muon candidate (highest muon-BDT score among qualifying tracks), exactly one contained pion candidate (LLR, MIP-BDT, pion-BDT, Bragg- π), three-plane hits, shower veto, $\theta_{\mu\pi} < 2.6$ rad, non-proton multiplicity < 4 and < 5 tracks.

Final cut	Signal	ν bkg	EXT	Pred.	Pur.
FHC	943	637	30	1 612	58.5%
RHC	1 015	720	23	1 759	57.7%
Comb.	1 959	1 357	53	3 372	58.1%

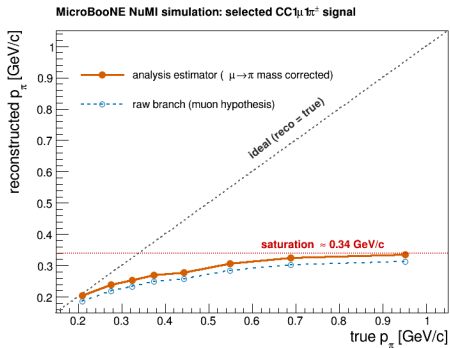
Efficiency 17.4–17.6% in the measured phase space; largest background: CC0 π with a proton taken as the pion.

Normalisation (details in backup): 8.86 (FHC) and 11.08×10^{20} POT (RHC), each run scaled by its own POT; per-configuration new-G4 flux; beam-off matched by run period and scaled by gates (1.9% of the sample).

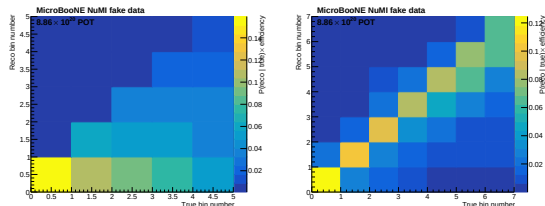


FHC cut-flow, per-run POT-scaled, full CV weight.

The pion momentum sets the scope



Reconstructed p_π (range under the muon hypothesis, mass-corrected) saturates near 0.34 GeV/c: pions interact before stopping.



Smearacceptance $S = \varepsilon M$: five-bin p_π (diagnostic, left) vs. p_μ (right).

- ▶ every estimator tried (golden-pion tag, MCS, calorimetry, regression, daughter recovery): a physics limit, not an algorithmic one
- ▶ p_π released as **two bins**: $[0.175, 0.205]$ and > 0.205 GeV/c, migration diagonals 91% / 76%
- ▶ proton-tagged $W_{\pi p}$ shape **withdrawn**; the sample supports an integrated cross section

Extraction: Wiener-SVD and what its integral means

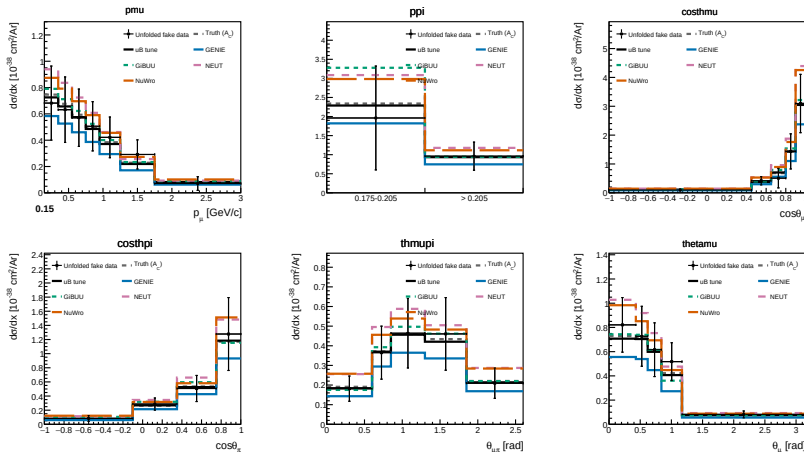
- ▶ $n^{\text{reco}} = S N^{\text{true}} + b$ with $S = \varepsilon M$ inverted once (no extra $1/\varepsilon$)
- ▶ Wiener-SVD (XSECANALYZER), second-derivative regularisation aware of the non-uniform bin widths
- ▶ the result is $A_C \times_{\text{true}}$; every prediction is smeared by the released A_C before comparison
- ▶ A_C is *not* norm preserving: the integral of a differential result is observable-dependent ($0.66\text{--}0.96 \times 10^{-38} \text{ cm}^2/\text{Ar}$); D'Agostini gives a flat, $\sim 30\%$ higher integral
- ▶ the physical total is the cut-and-count value

Total flux-averaged cross section ($10^{-38} \text{ cm}^2/\text{Ar}$)

	inclusive	proton-tagged
FHC	1.03 ± 0.42	0.63 ± 0.32
RHC	0.93 ± 0.35	0.55 ± 0.30
Comb.	0.98 ± 0.34	0.59 ± 0.28

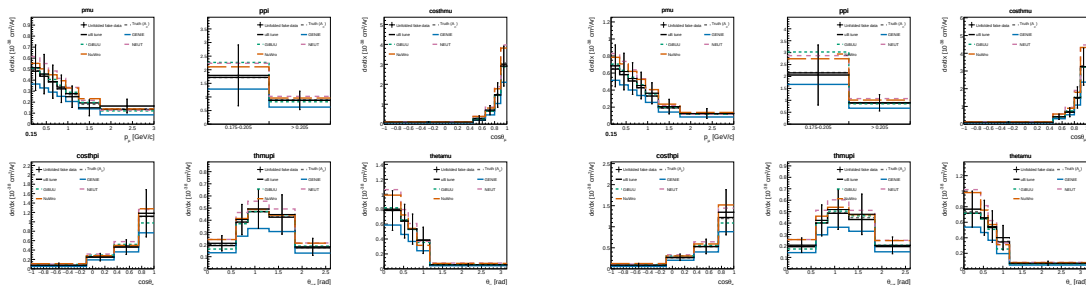
On fake data this *is* the injected tune (closure = 1 by construction); the uncertainty is what real data will carry. RHC/FHC = 0.90. GENIE v3 0.5–0.6 σ below, GiBUU/NEUT/NuWro within 0.7 σ .

Differential results, FHC



Unfolded fake data, A_C -smeared truth, tune and the four generators, all in the A_C -smeared space. Generators differ by normalisation ($\times 1.6$, GENIE to NEUT); shapes agree within the uncertainties. Full covariance: flux, cross section, re-interaction, detector, POT, targets, statistics.

RHC and combined

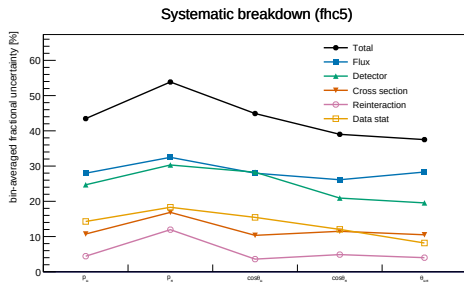


RHC

Combined FHC+RHC

Same 600 PPFX universes index-matched across both horn modes (rigorous flux correlation); native Run-4 RHC detector variations; each mode normalised to its own exposure and flux.

Uncertainties



Bin-averaged fractional uncertainty, inclusive (%):

Source	range	
Flux (PPFX)	24–33	dominant
Detector	16–35	~ flux for several obs.
Cross section	9–17	
Re-interaction	4–12	
Data statistics	6–18	
Total	32–54	

- ▶ the $\sim 17\%$ reco-level flux is amplified by the unfolding operator; the lever is response conditioning, not the flux model
- ▶ all 51 extractions carry a full covariance including detector variations; A_C and covariances released as text

Validation, and what the review found

Validation

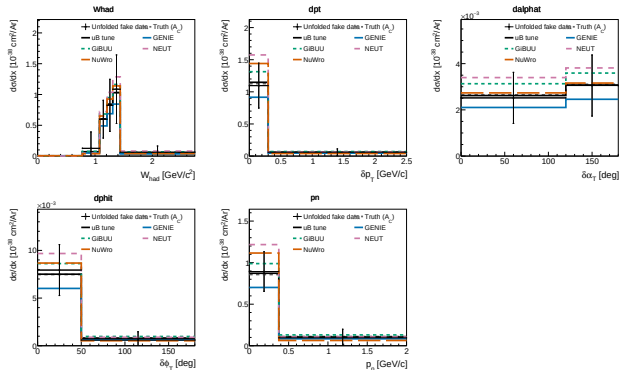
- ▶ fake-data closure: unfolded/realised truth = 1.018 on average over the 18 inclusive extractions; all $p > 0.9$
- ▶ 32-throw ensemble: pull widths 1.04 (p_μ), 0.96 ($\cos\theta_\mu$), 68/95% coverage recovered; 2.5% negative bias stated, not corrected
- ▶ clean-cache rebuild of all 51 extractions is bit-identical
- ▶ five automated guards: normalisation manifest, systematics/file coupling, staleness, table consistency, blinding guard

Corrected since August (all in the change log)

- ▶ beam-off scaled by an *event* count instead of gates: EXT was $9.7\times$ too large ($128 \rightarrow 30$ FHC events); now per run period
- ▶ fake-data truth carried the CV weight twice (1.6% on closure)
- ▶ PPFX CV weight missing from count-level tables (3–6%)
- ▶ one safe-weight rule everywhere; fake data re-thrown with it (1%)
- ▶ A_C applied twice to models; TKI computed about detector z (both withdrawn and redone)

None of these touched the real-data path; several would have biased a real-data result.

Proton-tagged subsample

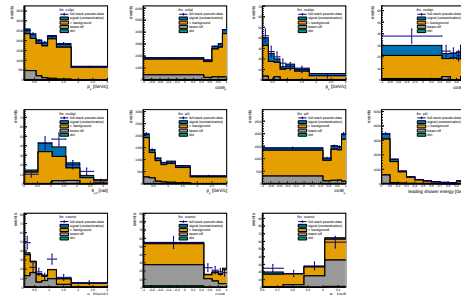


- ▶ inclusive selection + a contained proton $> 0.3 \text{ GeV}/c$; efficiency 11.4%, purity $\sim 50\%$; strong cosmic rejector
- ▶ TKI (δp_T , $\delta\alpha_T$, $\delta\phi_T$, p_n) in two bins, beam frame; W_{had} in six
- ▶ single-nucleon STV prescription applied to the πp system: p_n is a proxy (stated limitation)
- ▶ generators: normalisation spread only; no FSI discrimination at this exposure
- ▶ headline: integrated total 0.63 ± 0.31 (FHC)

Control regions

Region	Inversion	Role; $T = N^{\text{SR}}/N^{\text{CR}}$ of the target class
CC0 π	$n_\pi = 0$	tests 0 π feed-in; $T = 0.03$, $\delta T/T = 31\%$ (detector-limited)
multi- π π^0	$n_\pi \geq 2$ shower veto fails	diagnostic only; $T = 2.5$ π^0 leakage; $T = 0.02$, $\delta T/T = 15\%$: strongest handle
cosmic	$\theta_{\mu\pi} > 2.6$ rad	beam-off norm. and shape; 41% EXT purity, no fitted correction

- ▶ disjoint from the signal region event by event; multi- π and π^0 overlap ($\sim 40\%$ of multi- π)
- ▶ pseudo-data/prediction 0.99–1.10 everywhere; full-stack test (MC+EXT+dirt) at unity
- ▶ per-region systematics incl. detector variations; transfer factors as correlated ratios (flux cancels to 2–3%)



Frozen distributions and binning, FHC, full-stack pseudo-data in place of beam-on points.

Request to the collaboration

Open the four inclusive control regions; the signal region and the proton-tagged regions stay blind

Frozen before opening: selections at branch level, run-dependent normalisation, beam-off scaling, displayed observables and binning, covariance, goodness-of-fit and the permitted responses.

Procedure (to be ratified)

- ▶ FHC and RHC separately, combined as cross-check; data vs. MC+EXT+dirt
- ▶ pass: normalisation within 2σ (pre-fit predictive uncertainty) and shape $p > 0.05$
- ▶ allowed: no change; added uncertainty; one-parameter normalisation of the target class (purity $> 50\%$, LR test, transfer uncertainty added); else model revision and renewed approval
- ▶ forbidden: any signal cut or binning change; generator choice after inspection; bin-by-bin reweighting

Prerequisites

- ✓ tables from one release, checked automatically
- ✓ definitions, overlap = 0, plots and binning frozen
- ✓ EXT gates released per run; pairings asserted
- ✓ real beam-on files refused without XSEC.UNBLIND=1
- ✓ full-stack closure; region systematics incl. detector; transfer uncertainties
- ☐ ratification of the procedure

Before the signal region: PPFX-corrected flux constants confirmed; discrepancy treatment; clean-cache reproduction.

Summary

- ▶ A blind, fully validated extraction of $\nu_\mu + \bar{\nu}_\mu$ CC1 $\pi^\pm$ cross sections on argon in the NuMI beam: total and six differential distributions in FHC, RHC and combined, with full covariances and released A_C matrices.
- ▶ Two physics limits set the scope: the pion range saturates (two-bin p_π ; no $W_{\pi p}$ shape), and A_C is not norm preserving (the total is cut-and-count).
- ▶ Fake-data totals $1.03/0.93/0.98 \times 10^{-38} \text{ cm}^2/\text{Ar}$ with 35–41% uncertainty, flux- and detector-limited.
- ▶ A sequence of normalisation and weighting defects was found by review and fixed; automated guards now stop each of them from recurring.
- ▶ We ask to open the four inclusive control regions under the frozen procedure.

Documents: analysis note (Draft 1.3), technical supplement, change log; data release in `report/data_release/`.

Backup

Backup: exposure and normalisation

Configuration	Runs	POT (10^{20})
FHC	1, 2, 4, 5	8.857
RHC	1, 2, 3, 4a–c	11.082
Combined	all (14 overlay files)	19.939

	FHC	RHC
Run 1	3.283	0.605
Run 2	1.268	2.591
Run 3	–	5.003
Run 4	2.075	2.883
Run 5	2.231	–

Beam-on POT per run, 10^{20} .

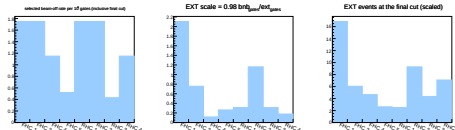
- ▶ every MC file scaled by $P_{\text{data}}(\text{run})/P_{\text{MC}}(\text{file})$; multi-file runs normalised by the sum of distinct file POT
- ▶ combined configuration = per-mode data exposures, one universe chain (600 PPFX universes index-matched across FHC and RHC)
- ▶ integrated flux (new-G4, $E_\nu > 60$ MeV): FHC 6.81, RHC 6.45, combined $6.61 \times 10^{-10} \text{ cm}^{-2}/\text{POT}$ (exposure-weighted); PPFX-corrected constants to be confirmed before unblinding
- ▶ $N_{\text{Ar}} = 8.71 \times 10^{29}$ in the fiducial volume (dead z slab removed)
- ▶ CV weight = tune $\times$ PPFX $\times$ normalisation; safe rule: non-finite, negative or $> 30 \rightarrow 1$, zero kept. GENIE universes carry the tune; PPFX universes are absolute; re-interaction universes relative

Backup: beam-off per run period

Mode	Run	Beam-on gates	EXT files	EXT gates	scale
FHC	1	9 846 635	run1	4 582 248	2.106
FHC	2	3 535 129	run1 (stand-in)	4 582 248	0.756
FHC	4	4 131 149	run4a-d	34 831 149	0.116
FHC	5	5 154 196	run5	19 256 341	0.262
RHC	1	1 458 253	run1	4 582 248	0.312
RHC	2	5 422 907	run1 (stand-in)	4 582 248	1.160
RHC	3	10 349 610	run3b	32 649 129	0.311
RHC	4	6 304 167	run4a-d	34 831 149	0.177

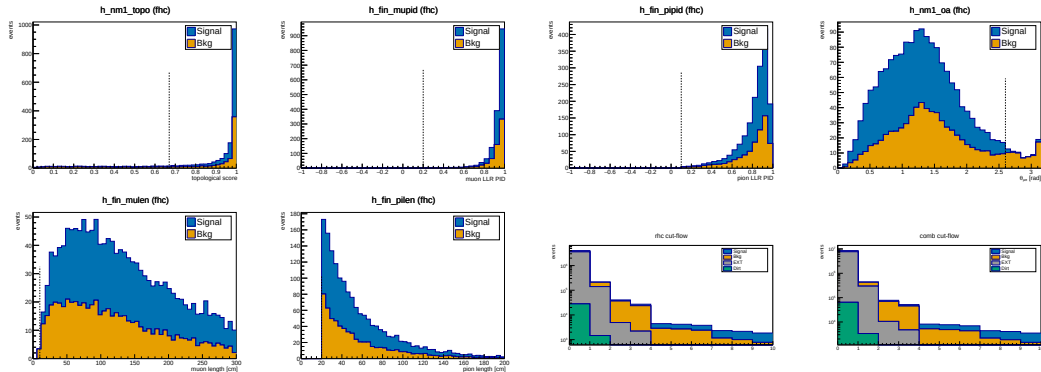
scale = $0.98 \times \text{beam-on gates} / \sum \text{EXT gates}$.

- ▶ cosmic-only data: the horn label is irrelevant, only the run period (CRT, detector conditions) matters
- ▶ no Run-2 beam-off exists; Run 1 is the stand-in (same pre-CRT conditions); Run 4 pools 4a-4d
- ▶ final-cut EXT: 30.2 (FHC), 23.2 (RHC), 53.4 (combined) events, 1.9% of the sample
- ▶ the previous release scaled by an *event* count: $9.7\times$ too much EXT



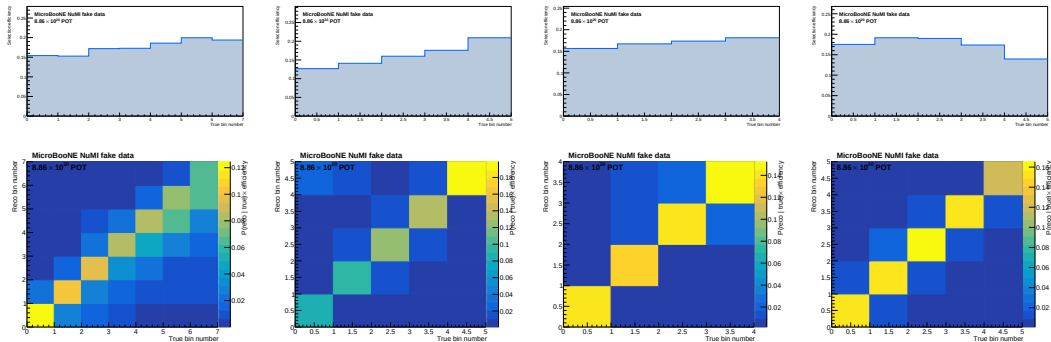
Selected cosmic rate per million gates: pre-CRT
Run 1 is $3-4\times$ the CRT-era runs, as expected.

Backup: selection variables (FHC)



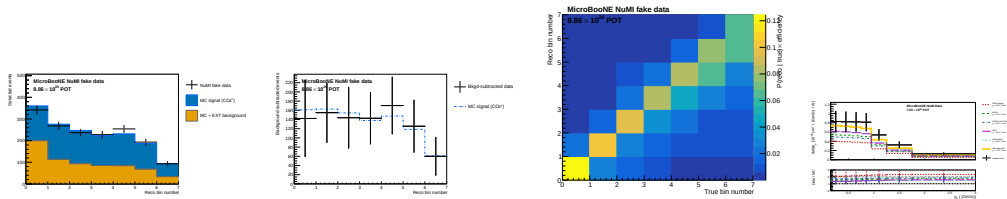
Top: topological score ($N - 1$), muon-BDT score, pion-BDT score, opening angle ($N - 1$). Bottom: muon and pion track lengths at the final cut; RHC and combined cut-flows. All per-run POT-scaled, full CV weight, fake data as points.

Backup: efficiency and response



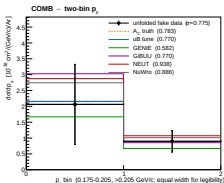
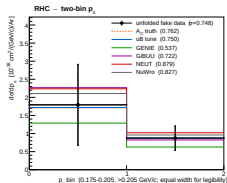
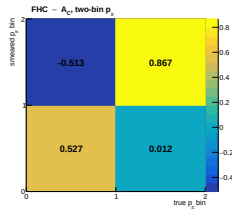
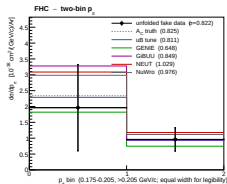
Top: selection efficiency vs. true variable (FHC). Bottom: smear acceptance $S = \varepsilon M$ used in $n^{\text{reco}} = S N^{\text{true}} + b$; top true bins are open-ended. Efficiency is folded into S once; no separate $1/\varepsilon$ division.

Backup: from reco distributions to the unfolded result (p_μ , FHC)



- ▶ left to right: reco-level stack with fake data; background-subtracted; A_C (additional smearing) matrix; unfolded cross section with the A_C -smeared truth and models
- ▶ the Wiener-SVD regulariser is a second-derivative operator in bin index, corrected for the non-uniform bin widths (`fw_regmatrix`)
- ▶ the released products per extraction: A_C , unfolded values, full covariance, and the smeared predictions

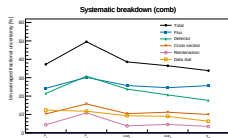
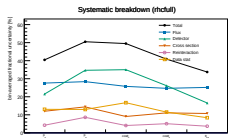
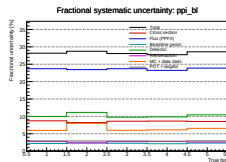
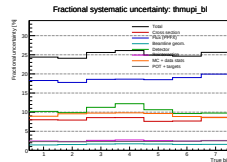
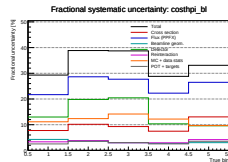
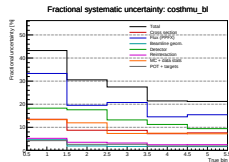
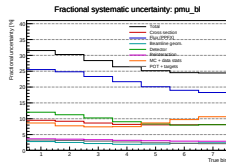
Backup: two-bin pion momentum



Two-bin $d\sigma/dp_\pi$ and its A_C : FHC (top), RHC and combined (bottom).

- ▶ range-based p_π under the muon hypothesis with the mass correction applied in the bin configuration; saturates at ~ 0.34 GeV/c because most pions interact before stopping
- ▶ five-bin diagnostic: top true bin only 6% diagonal (criterion > 0.68)
- ▶ estimators tried and rejected: golden-pion tagging ($40 \rightarrow 53\%$ golden at 87% efficiency, insufficient), MCS, calorimetric, regression on track variables, daughter recovery
- ▶ two bins $[0.175, 0.205]$ and > 0.205 GeV/c: diagonals 91% / 76%, released for all three configurations and the proton-tagged subsample
- ▶ same limit removes the proton-tagged $W_{\pi p}$ shape (pion momentum enters directly)

Backup: uncertainty breakdown by observable and configuration



Flux: 600 PPFX universes.
Cross section: GENIE multisim
+ unisims.
Re-interaction: Geant4
reweight.
Detector: Run-4 variation
samples, FHC and RHC na-
tively; proton-tagged included.
POT 2%, targets 1%, statistics
of MC and fake data.

Detector variations are
17–43% after unfolding: the
response of a $\sim 5\%$ reco-level
shift is amplified by the poorly
conditioned S .

Top: FHC by observable. Bottom: two-bin p_π , then RHC and combined (p_μ).

Backup: closure and ensemble tests

Fake-data closure (18 inclusive extractions)

- ▶ fake data: an external Poisson throw of the CV prediction per run, weights reset to one in both reco and truth
- ▶ unfolded / realised truth = 1.018 on average; χ^2 probabilities > 0.9 everywhere
- ▶ numuMC self-closure: unfolded/truth 0.94–1.06 for every observable, so the Wiener-SVD is unbiased on the injected model
- ▶ the A_C -smeared integral is observable-dependent by construction; D'Agostini gives a flat integral $\sim 30\%$ higher, iteration-stable

32-throw ensemble

- ▶ pull widths 1.04 (p_μ) and 0.96 ($\cos \theta_\mu$); 68% and 95% coverage recovered within the ensemble statistics
- ▶ 2.5% negative bias in the ensemble mean, stated and not corrected
- ▶ $\cos \theta_\mu$: the strongly suppressing A_C does not distort the uncertainties
- ▶ FHC θ_μ : A_C is rank-1 (all models share one shape after smearing), so θ_μ carries no shape information and is released as a normalisation check only

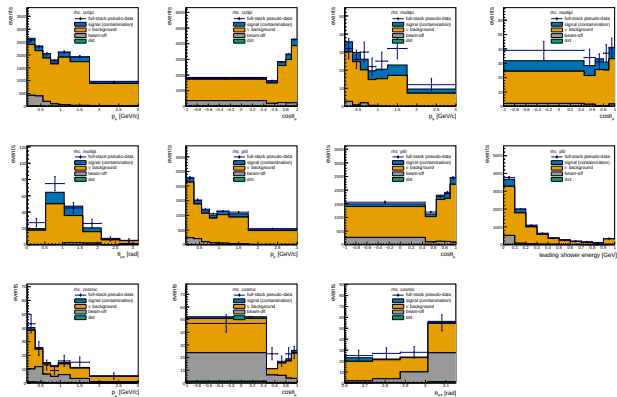
Reproducibility: clean-cache rebuild of all 51 extractions is bit-identical; the universe cache is keyed on the normalisation scheme and on the file list.

Backup: control-region transfer factors (FHC)

Class	SR	CC0 π	multi- π	π^0	cosmic	$T_{\text{CC0}\pi}$	T_{multi}	T_{π^0}	T_{cosmic}
signal (contamination)	943	1 124	29.7	662	6.0	0.84	32	1.43	158
ν_μ CC 0 π (track taken as pion)	244	8 781	2.6	1 011	23.2	0.03	95	0.24	11
ν_μ CC 1 $\pi^\pm$ out of phase space	52.7	629	2.7	361	14.8	0.08	19	0.15	3.57
ν_μ CC with a π^0	58.0	408	20.2	3 070	2.5	0.14	2.87	0.02	23
ν_μ CC $\geq 2\pi^\pm$, no π^0	145	222	57.8	658	3.9	0.65	2.51	0.22	37
out of fiducial volume	47.4	480	3.3	253	18.5	0.10	14	0.19	2.56
NC and ν_e CC	89.5	549	7.1	632	6.4	0.16	13	0.14	14
total ν -MC	1 581	12 192	124	6 646	75.3				

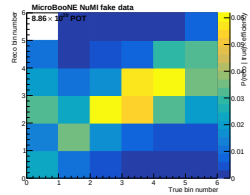
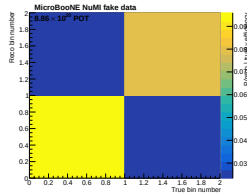
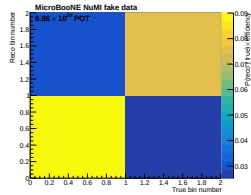
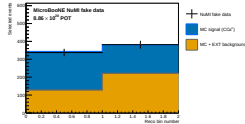
- ▶ $T = N^{\text{SR}}/N^{\text{CR}}$ per truth class; a normalisation change of the target class in a region propagates to the signal region by T
- ▶ CC0 π : $T = 0.03$, $\delta T/T = 31\%$ (detector-variation limited); π^0 : $T = 0.02$, $\delta T/T = 15\%$; flux cancels to 2–3% in the ratio
- ▶ the multi- π region has $T = 2.5$ and a 41% signal share: diagnostic only, no correction may be derived from it
- ▶ the cosmic region is 41% EXT with the rest dominated by CC0 π ; it tests the beam-off normalisation and shape, no fitted correction

Backup: control regions, RHC



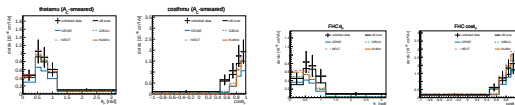
Frozen distributions (p_μ , $\cos\theta_\mu$, opening angle, shower energy) and binning for the four RHC regions; full-stack pseudo-data (MC+EXT+dirt, Poisson-thrown) in place of beam-on points. Pseudo-data/prediction 0.99–1.10.

Backup: proton-tagged kinematics



- ▶ TKI built in the NuMI beam frame from the μ and the leading proton with the pion added to the hadronic system; before the beam-frame fix δp_T was computed about detector z (0.74 vs. 0.26 GeV/c mean)
- ▶ two-bin releases for δp_T , $\delta \alpha_T$, $\delta \phi_T$, p_n ; six bins in W_{had} ; $W_{\pi p}$ withdrawn (pion momentum)
- ▶ all proton-tagged extractions carry detector variations (10–64%, median 24%)
- ▶ the two-bin shapes separate generators by normalisation only; FSI discrimination would need the full exposure and better p_π

Backup: angles in the beam frame



Left: θ_{μ} vs. $\cos\theta_{\mu}$ in the beam frame. Right: the two released θ_{μ} observables are the same data with different binning.

- ▶ beam direction: truth uses the per-event neutrino direction; reco uses target-to-vertex, with the convention of the NuMI ν_e analysis
- ▶ 64.6% of events changed $\cos\theta_{\mu}$ bin when the axis was corrected from detector z to the beam
- ▶ p_{μ} , p_{π} , $\theta_{\mu\pi}$ and the selection are frame-independent and were unaffected
- ▶ $\cos\theta_{\pi}$ coarsened $5 \rightarrow 4$ and $\theta_{\mu\pi}$ $7 \rightarrow 5$ bins to limit A_C smearing